

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/E_DEB_SAAWZe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3333, C2 = 3663, C3 = 2212)

text_priors <- '
Distrib (Fm,          TruncNormal,          0.1,          0.2, 0.01,          0.8);
Distrib (pAm,         TruncNormal_cv,        100,          0.2, 40,          2000);
Distrib (r_pAm_pM,    TruncNormal_cv,        0.8,          0.1, 0.1,          2);
Distrib (v,           TruncNormal_cv,        0.18505,       0.1, 0.001,        2);
Distrib (kap,         Uniform,                0,            1);
Distrib (kapA,        Uniform,                0.1,          0.9);
Distrib (Ehp,         TruncNormal_cv,        100,          0.2, 40,          300);
Distrib (E_coc,       TruncNormal_cv,        30,          0.2, 1,          300);
Distrib (rOM_ClxHorse, LogUniform,          0.00001,       0.3);
Distrib (z,           Uniform,                0,            1);
Distrib (dv,          TruncNormal_cv,        2,           0.5, 0,          10);
Distrib (wE,          LogUniform,          0.000000001,   0.1);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter, seeds)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

Fm.1. pAm.1. r_pAm_pM.1. v.1. kap.1. kapA.1.

Result_mode	0.0105876	216.63100	1.0019500	0.20850600	0.9585990	0.7732880
2.5%	0.0100549	40.13911	0.6027710	0.00612356	0.2333765	0.1044970
97.5%	0.5804150	242.47100	1.0325915	0.23204725	0.9799150	0.8538860
Min	0.0100002	40.00160	0.4703130	0.00545438	0.1767450	0.1000200
Max	0.7423990	263.15100	1.1395800	0.28614700	0.9851500	0.8902430
Result_mean	0.3135068	104.26736	0.8157606	0.07383602	0.5539962	0.3703895
Result_sd	0.2195978	86.06548	0.1220234	0.09480258	0.2986125	0.3129157
	Ehp.1.	E_coc.1.	rOM_ClxFhorse.1.		z.1.	dv.1.
Result_mode	110.03600	34.075100	0.0001186060	0.9970060000	1.715450	
2.5%	52.41770	24.678200	0.0000127354	0.0276685300	1.236640	
97.5%	127.57400	47.187600	0.1790550000	0.9985110000	4.666800	
Min	40.40340	18.374600	0.0000100601	0.0000284216	0.522040	
Max	165.13200	57.043900	0.2862270000	0.9999990000	5.729970	
Result_mean	89.16966	35.135813	0.0275510434	0.6256485389	2.883835	
Result_sd	19.55224	5.813617	0.0525423222	0.3435261425	1.126617	
	wE.1.	Sigma_W.1.				
Result_mode	1.337860e-09	0.20401900				
2.5%	1.443760e-09	0.18867300				
97.5%	1.589012e-06	0.26931485				
Min	1.008530e-09	0.17025000				
Max	4.221820e-06	5.72094000				
Result_mean	7.844752e-07	0.23114242				
Result_sd	5.152881e-07	0.05640154				

Number of observations, n = 277
 Number of parameters, k = 12
 Log-likelihood, LL = -132.7923
 BIC = 333.0728

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	3.67	8.14
E_coc.1.	1.34	1.97
Ehp.1.	1.20	1.57
Fm.1.	8.12	19.86
kap.1.	9.09	22.07
kapA.1.	19.72	38.20
pAm.1.	23.61	81.13
r_pAm_pM.1.	2.67	4.95
rOM_ClxFhorse.1.	2.12	10.82

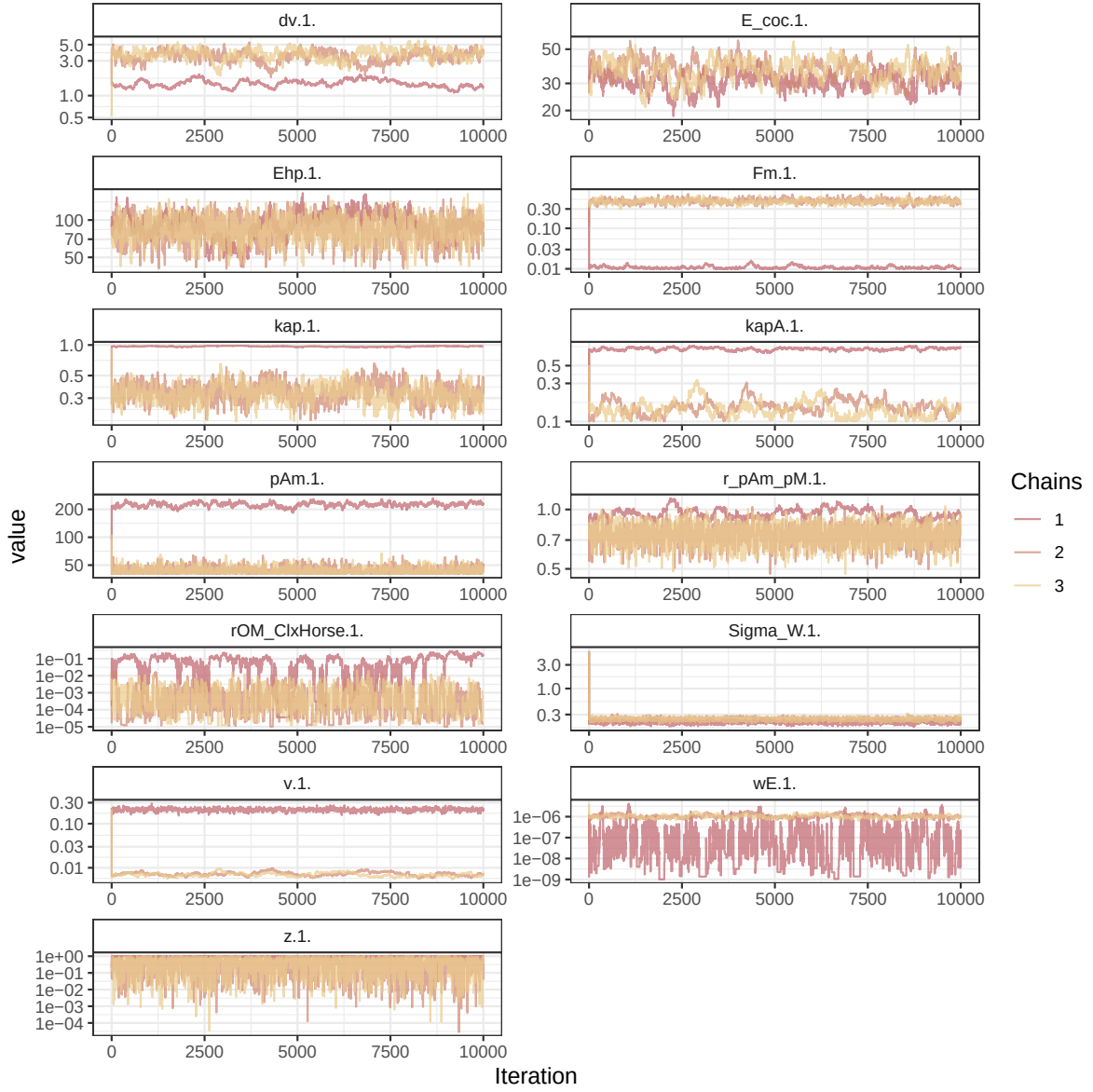


Figure 1: Parameters chains. Only the first iteration and the last `Nb_iter_kept` are represented.

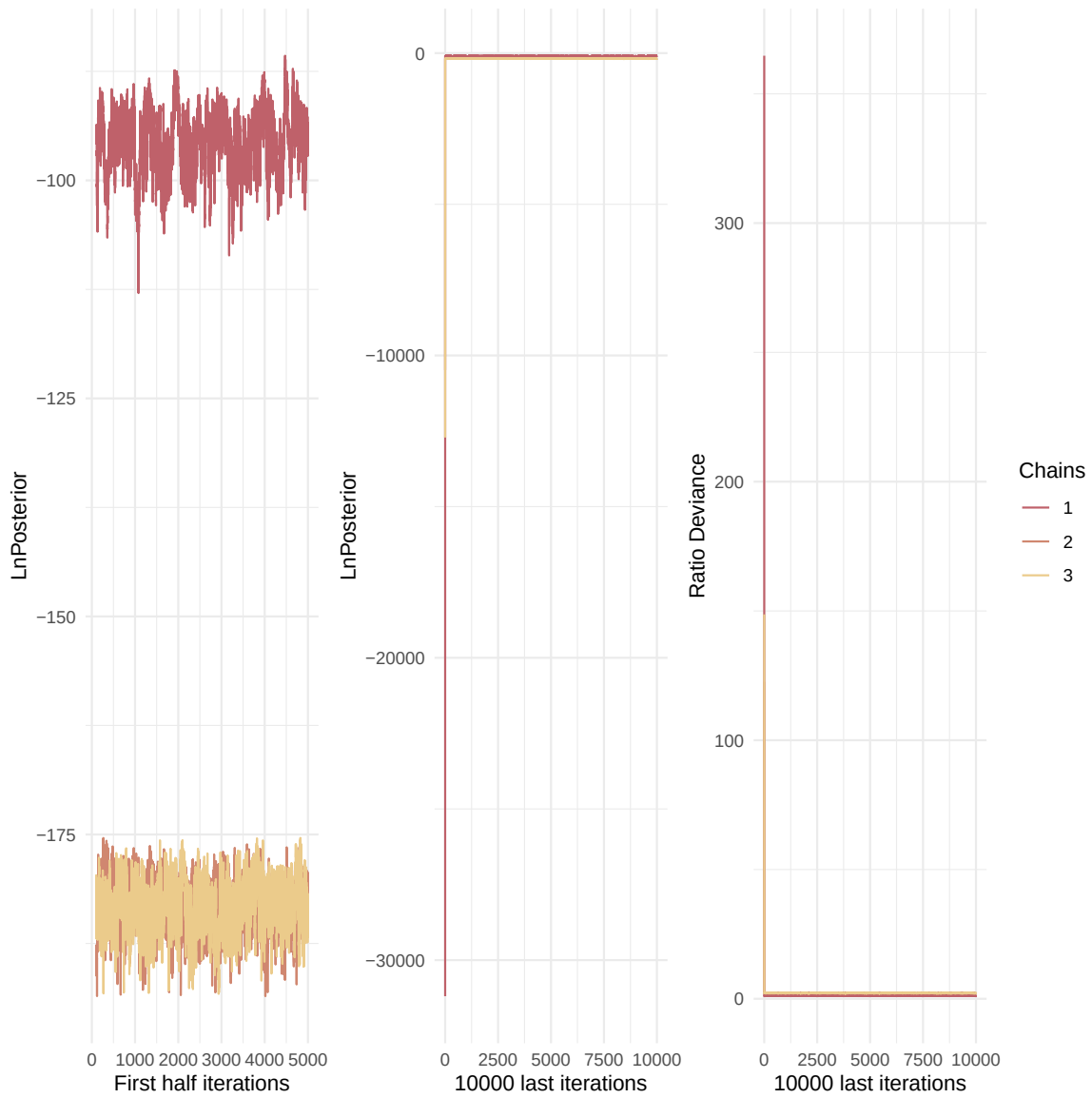


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

Sigma_W.1.	2.21	3.77
v.1.	17.76	109.13
wE.1.	2.52	6.31
z.1.	2.11	4.48

Multivariate psrf

25.4

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

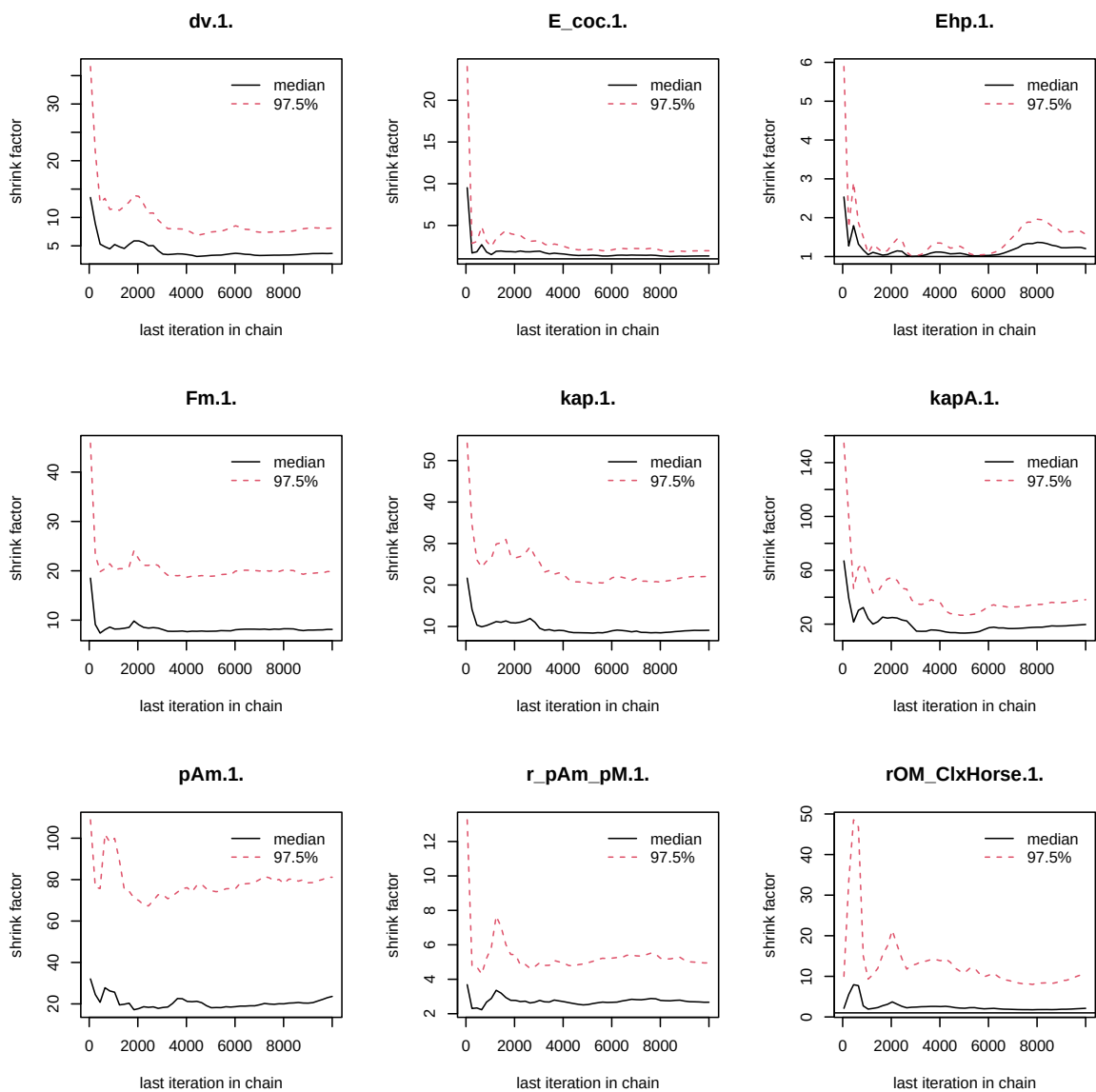


Figure 3: Chains convergence.

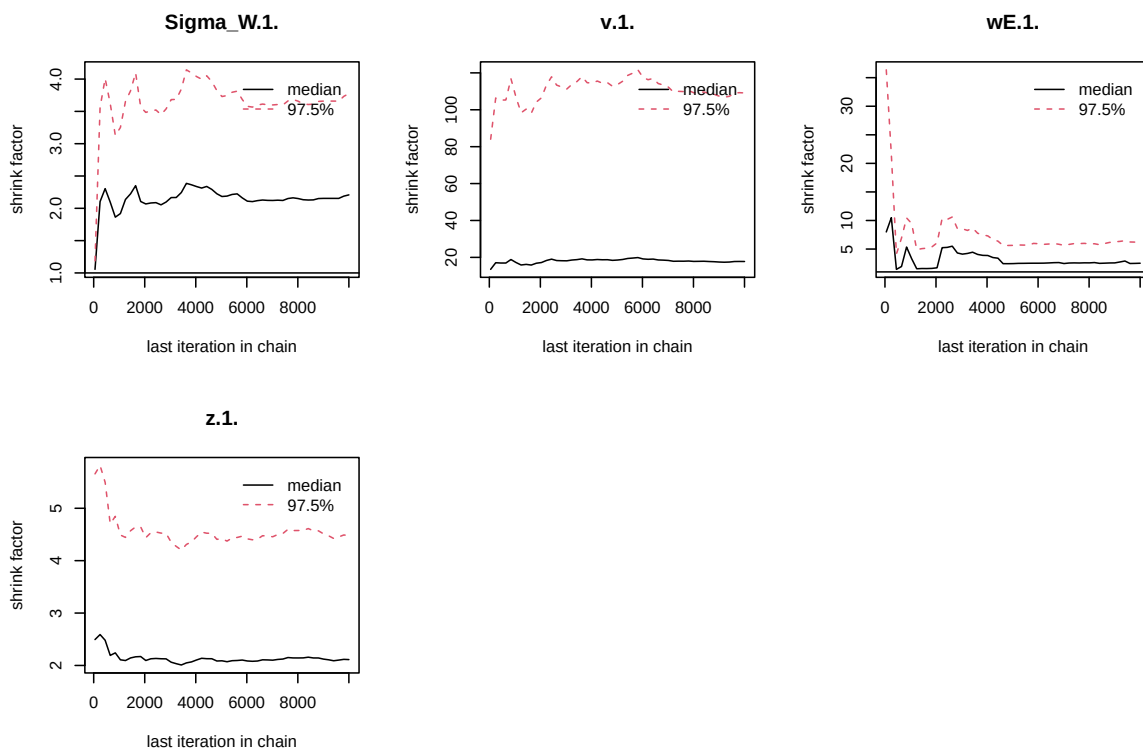


Figure 4: Chains convergence.

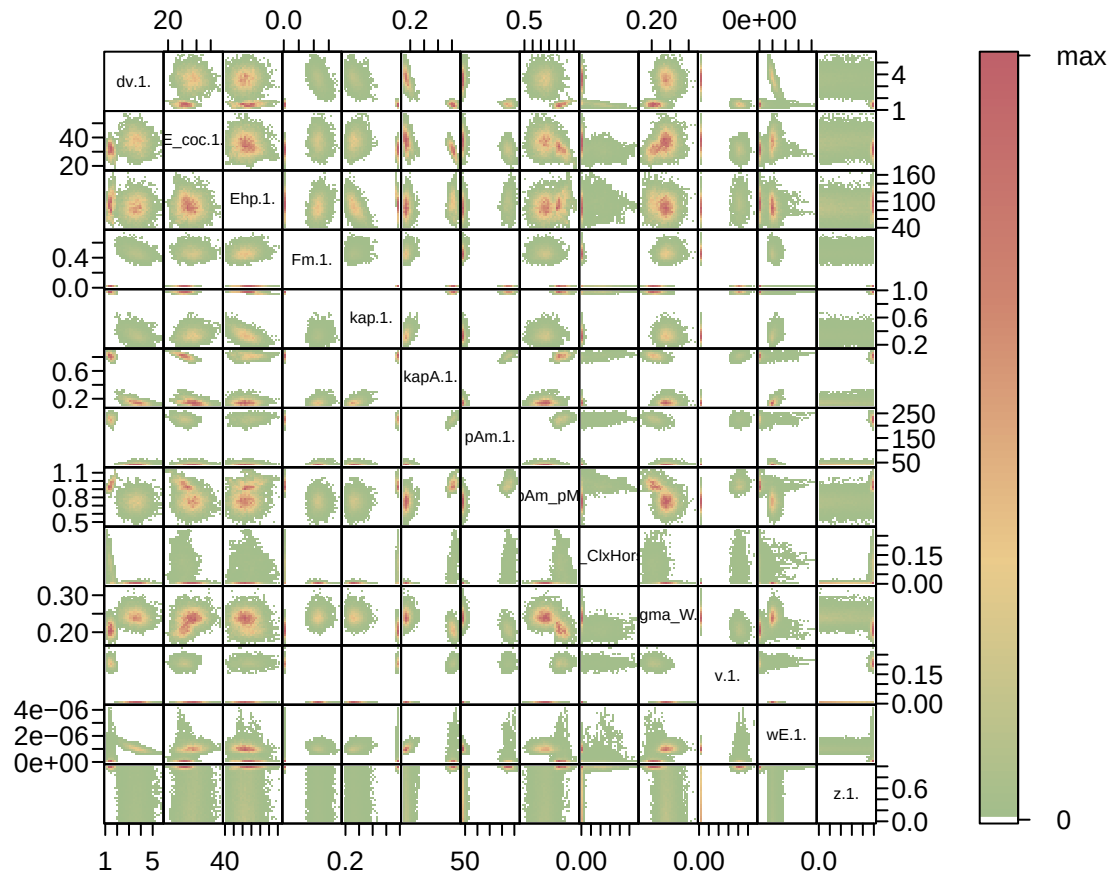


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

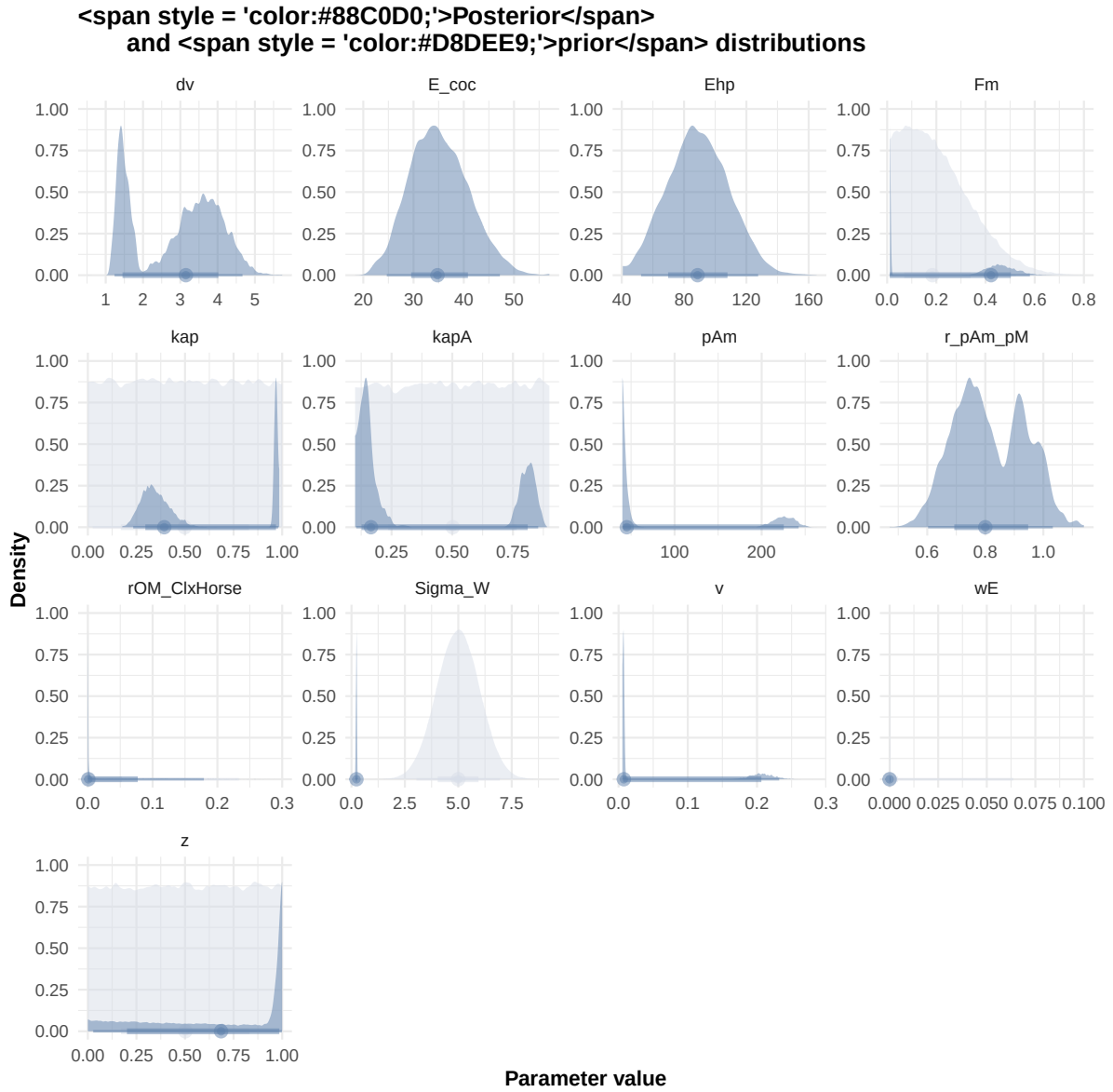


Figure 6: Distributions of the priors and the posteriors.

```

"Bart2020",
#"Gollot2026_lufa"
"Gollot2026_dens_D1",
"Gollot2026_dens_D2",
"Gollot2026_dens_D2b",
"Gollot2026_dens_D5",
"Gollot2026_dens_D7"
)

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\\\.1\\.\\.$|\\.1\\.\\.", "", param),
    line = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
  Length_struct,
  Reproduction,
  Energy,
  Organic_matter,"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_print)
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcali DEB_sim_full.in

#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),

```

```

    color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
              Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
  )
  v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
  #####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)

```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected relationship between variables
 i Row 1 of `x` matches multiple rows in `y`.
 i Row 1 of `y` matches multiple rows in `x`.
 i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

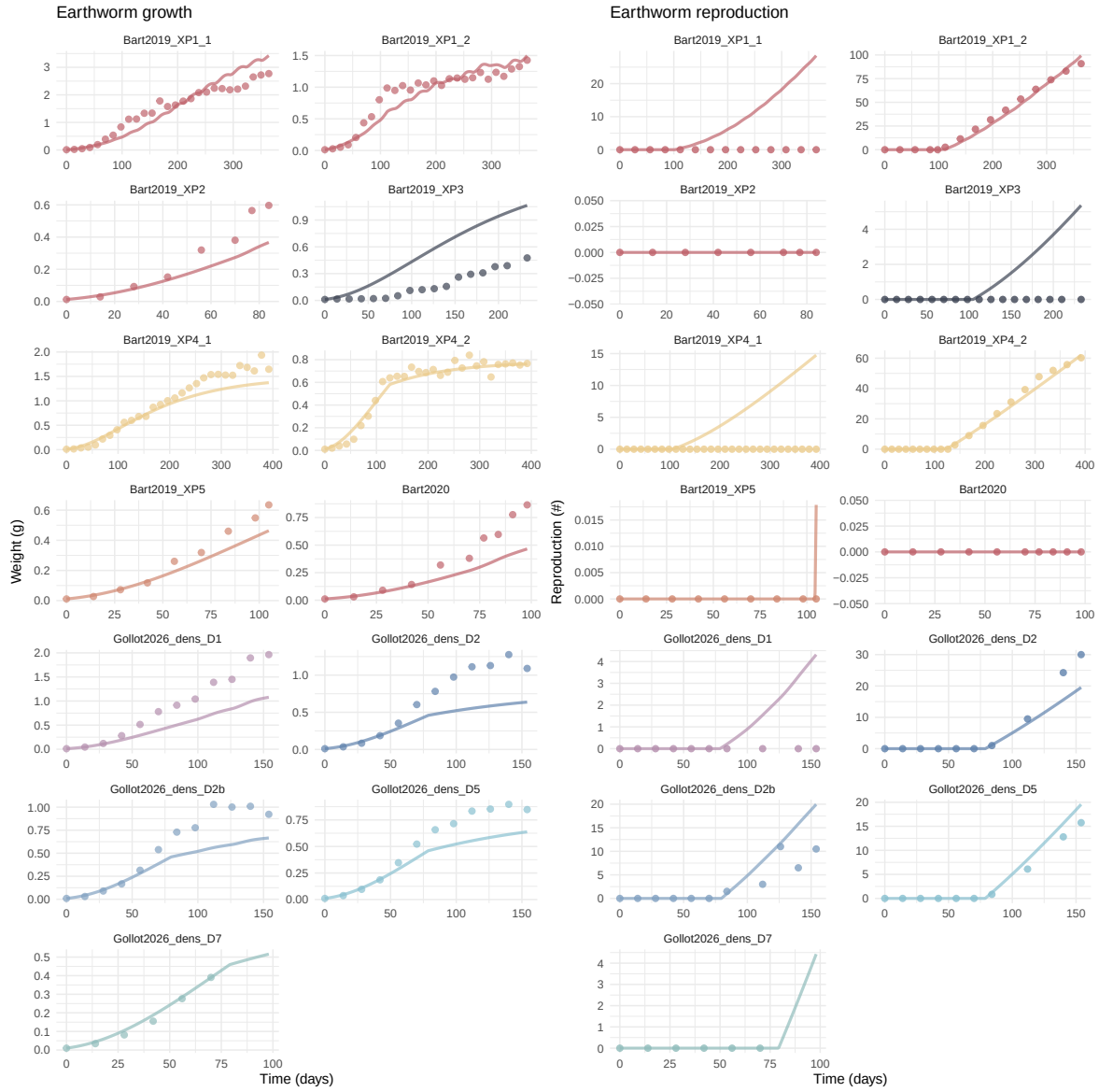
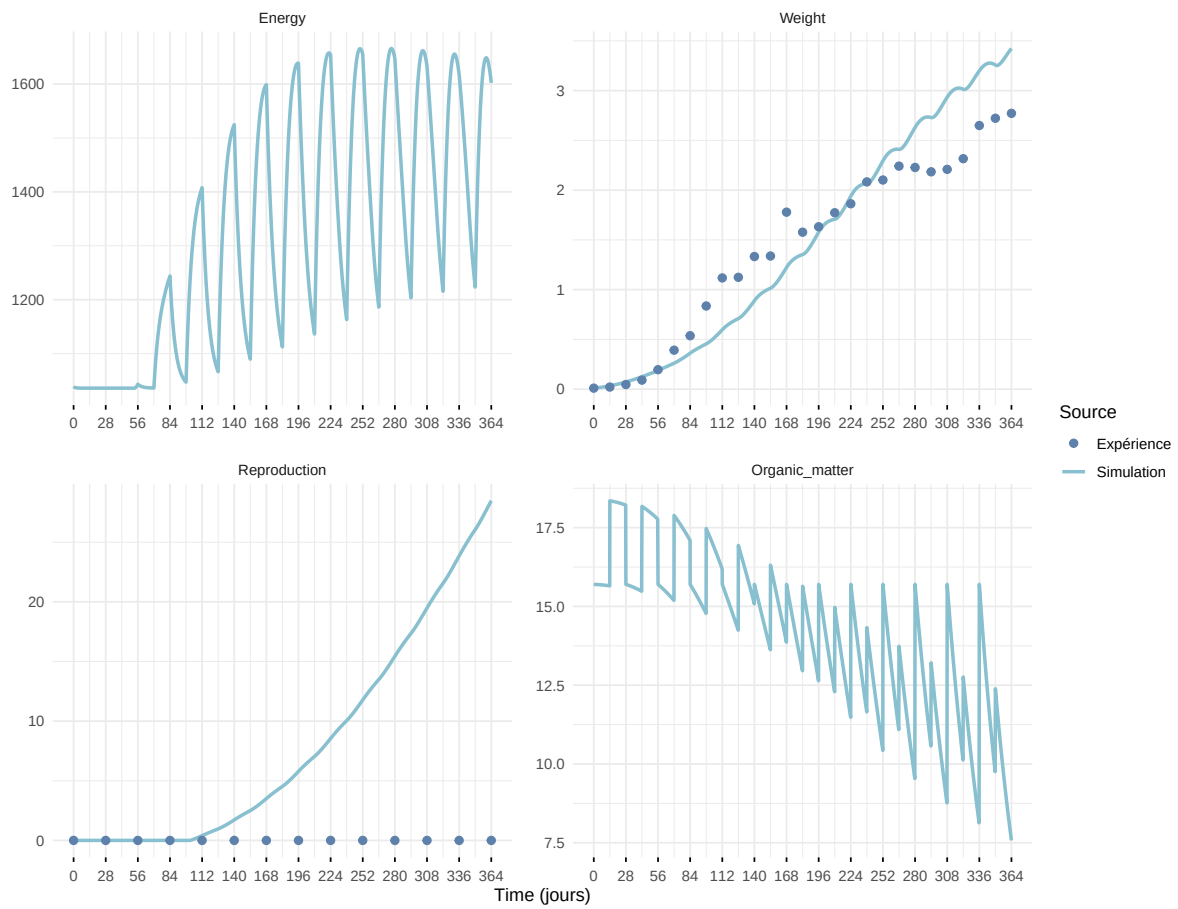
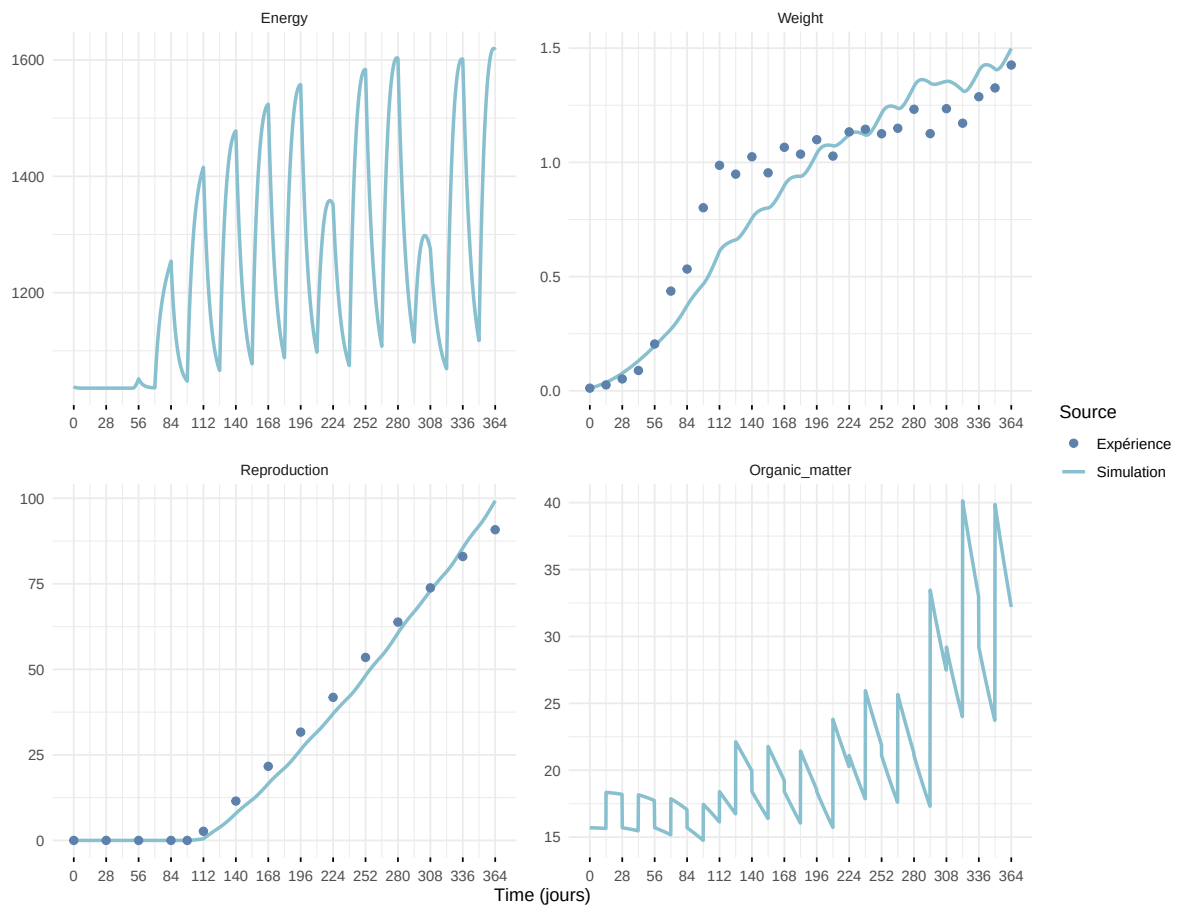


Figure 7: Simulations Weight and Reproduction.

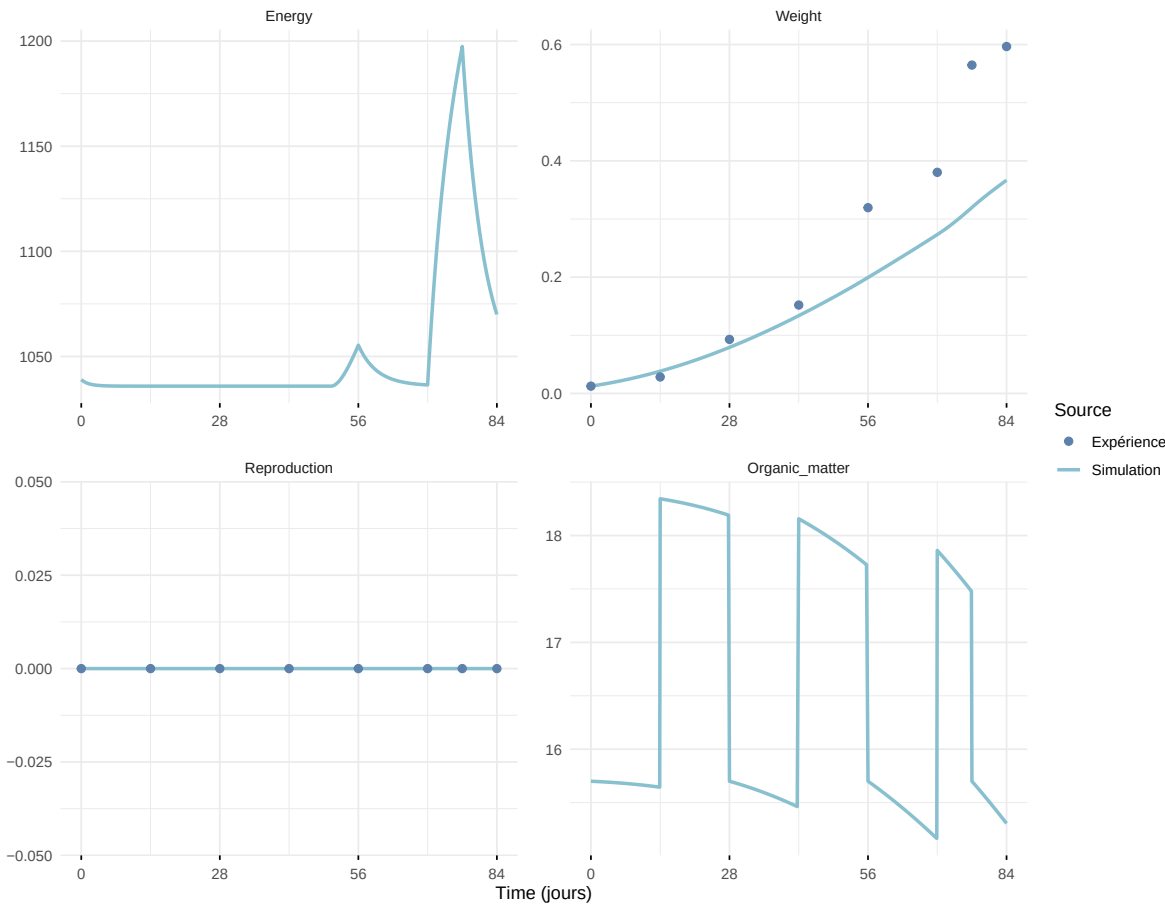
Expérience : Bart2019_XP1_1



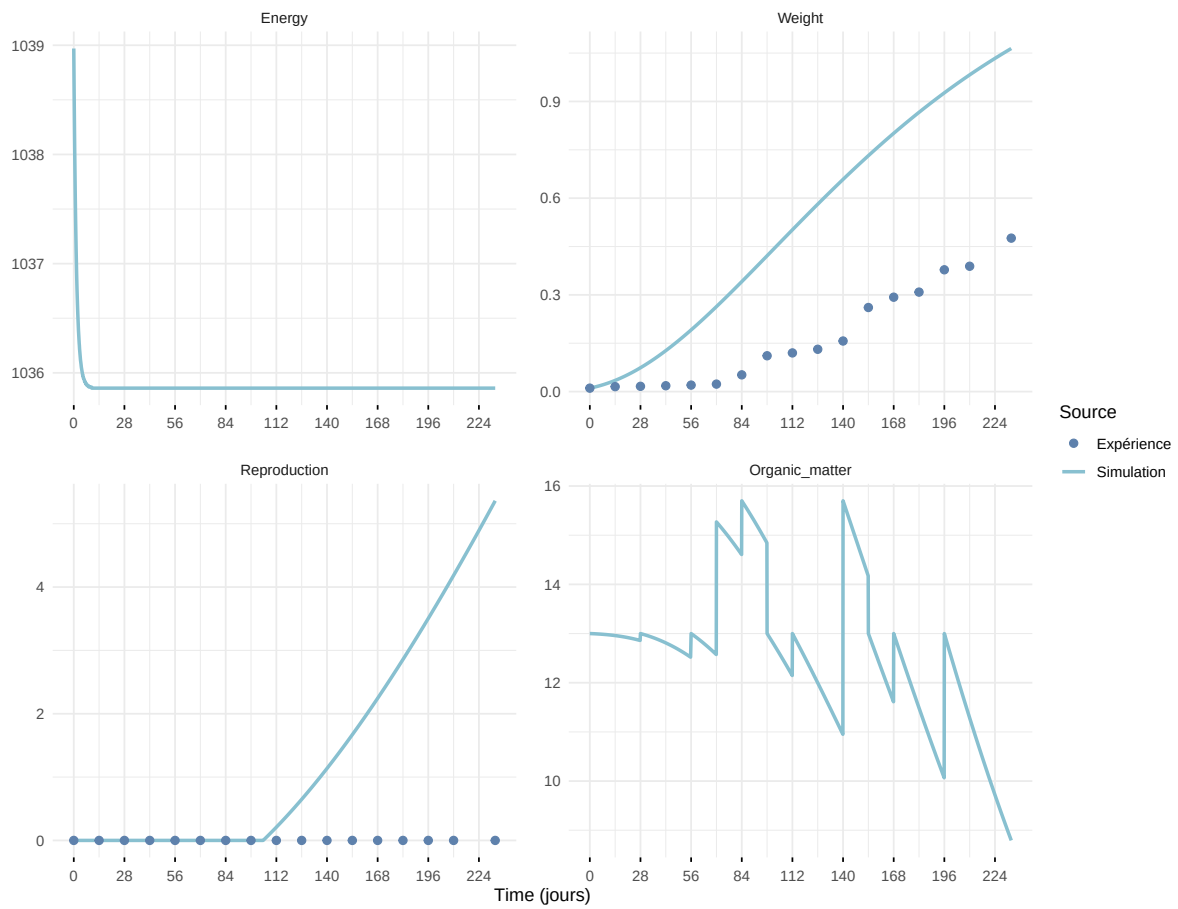
Expérience : Bart2019_XP1_2



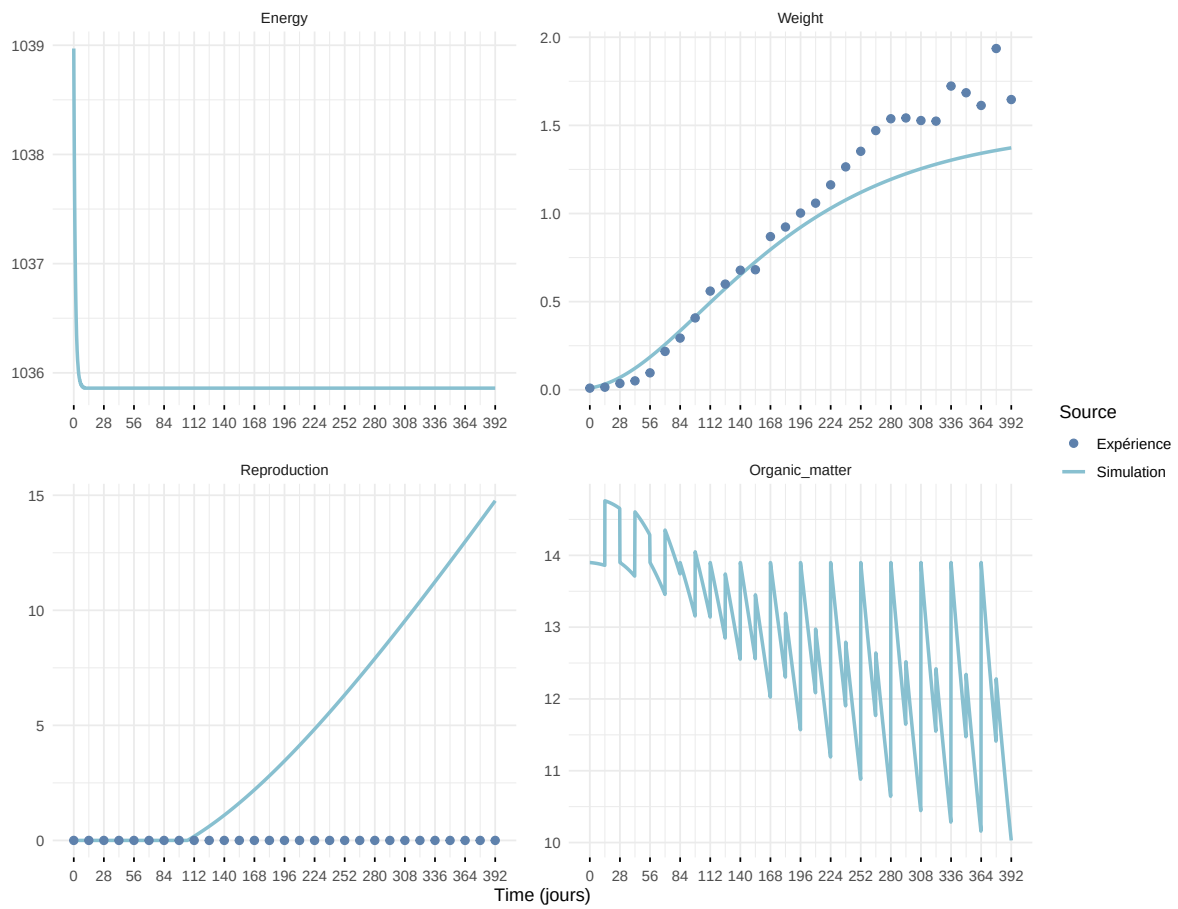
Expérience : Bart2019_XP2



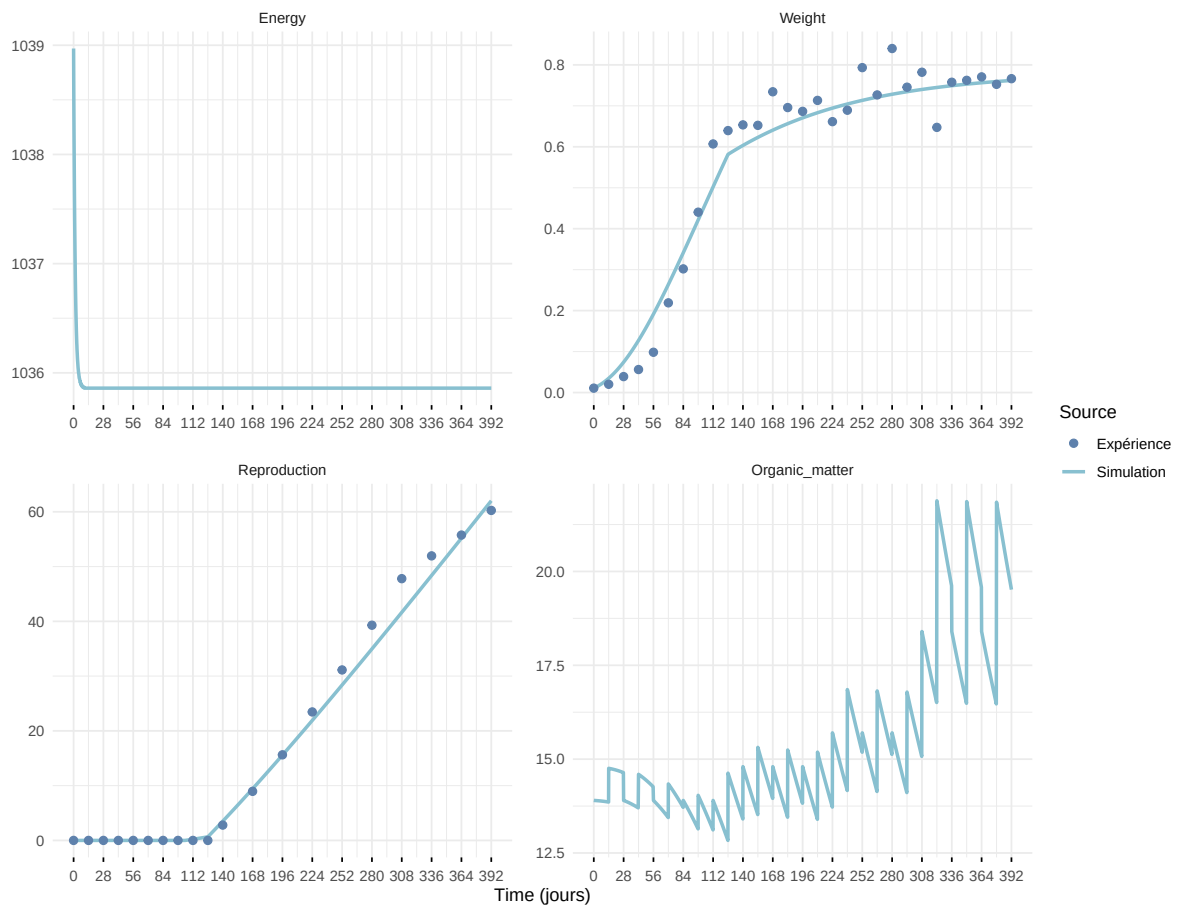
Expérience : Bart2019_XP3



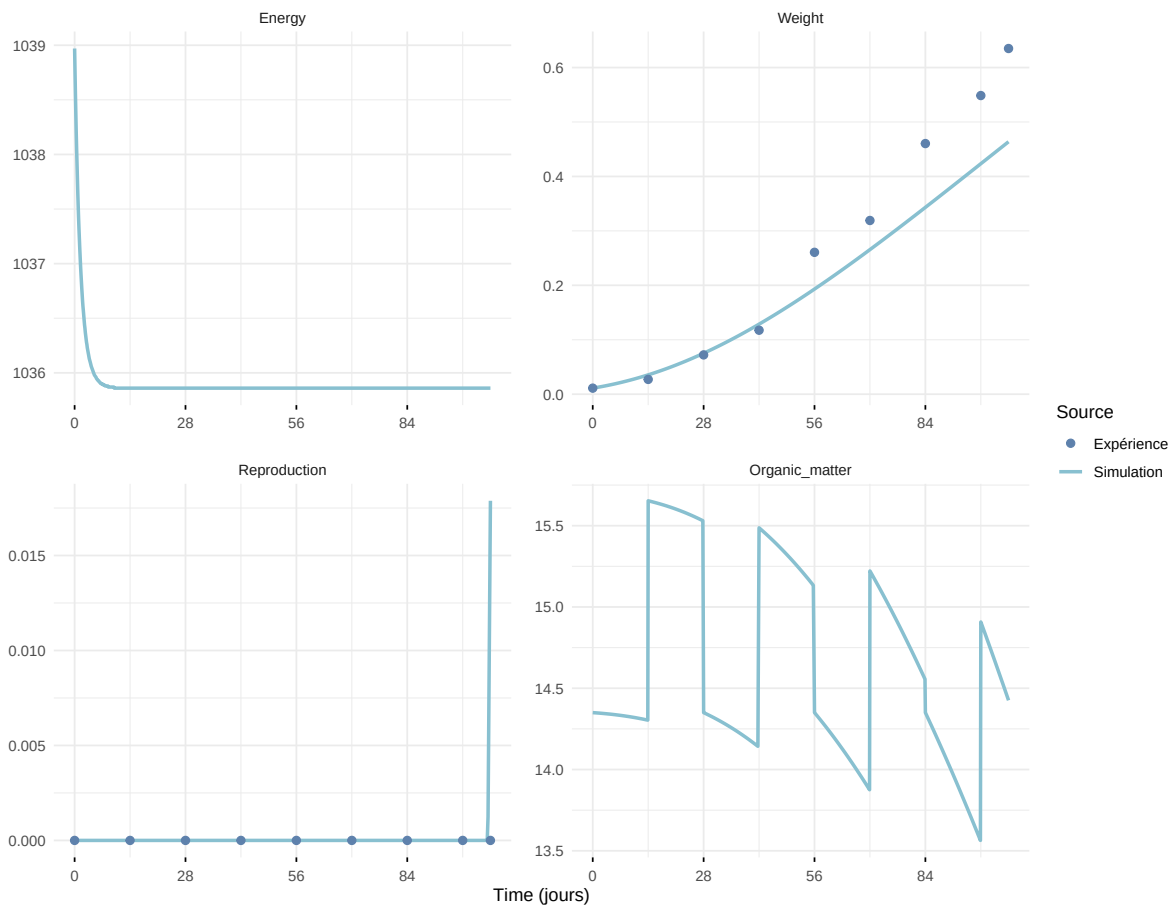
Expérience : Bart2019_XP4_1



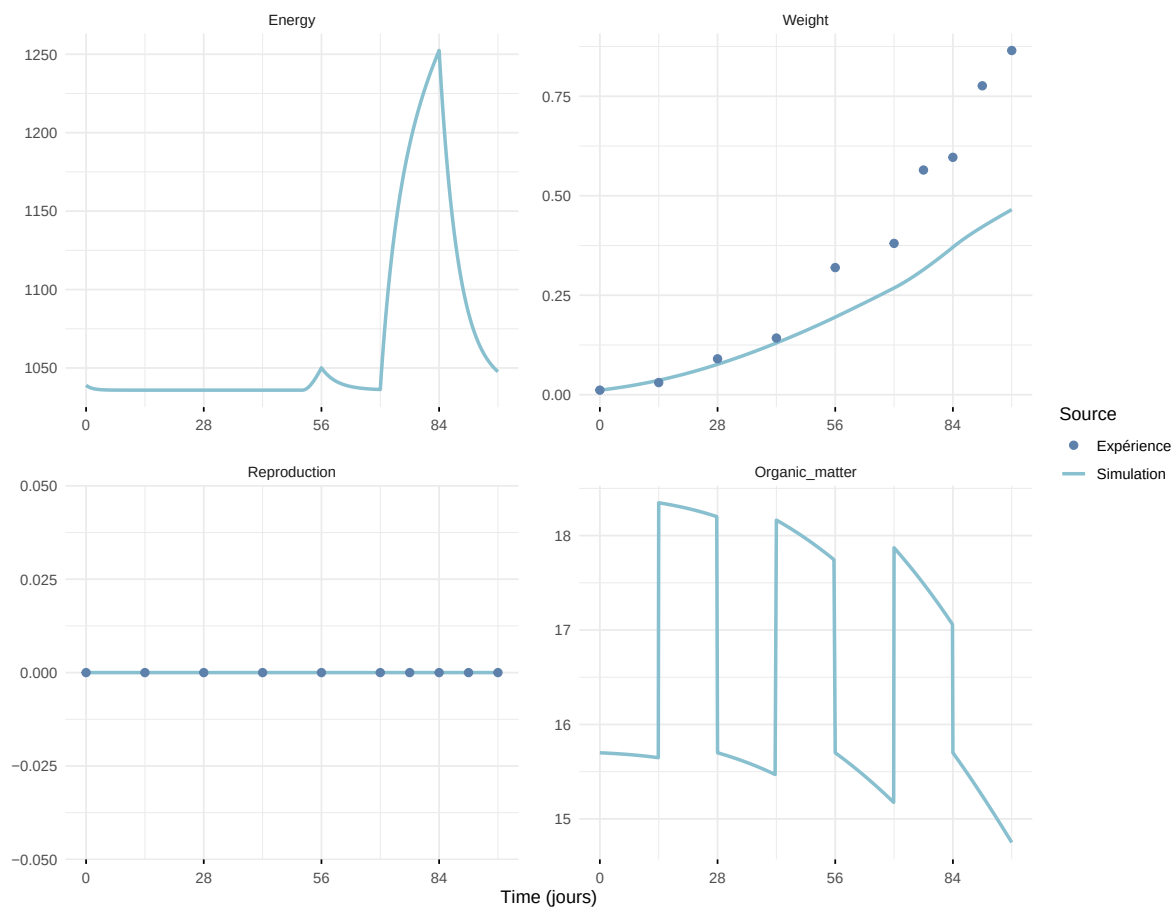
Expérience : Bart2019_XP4_2



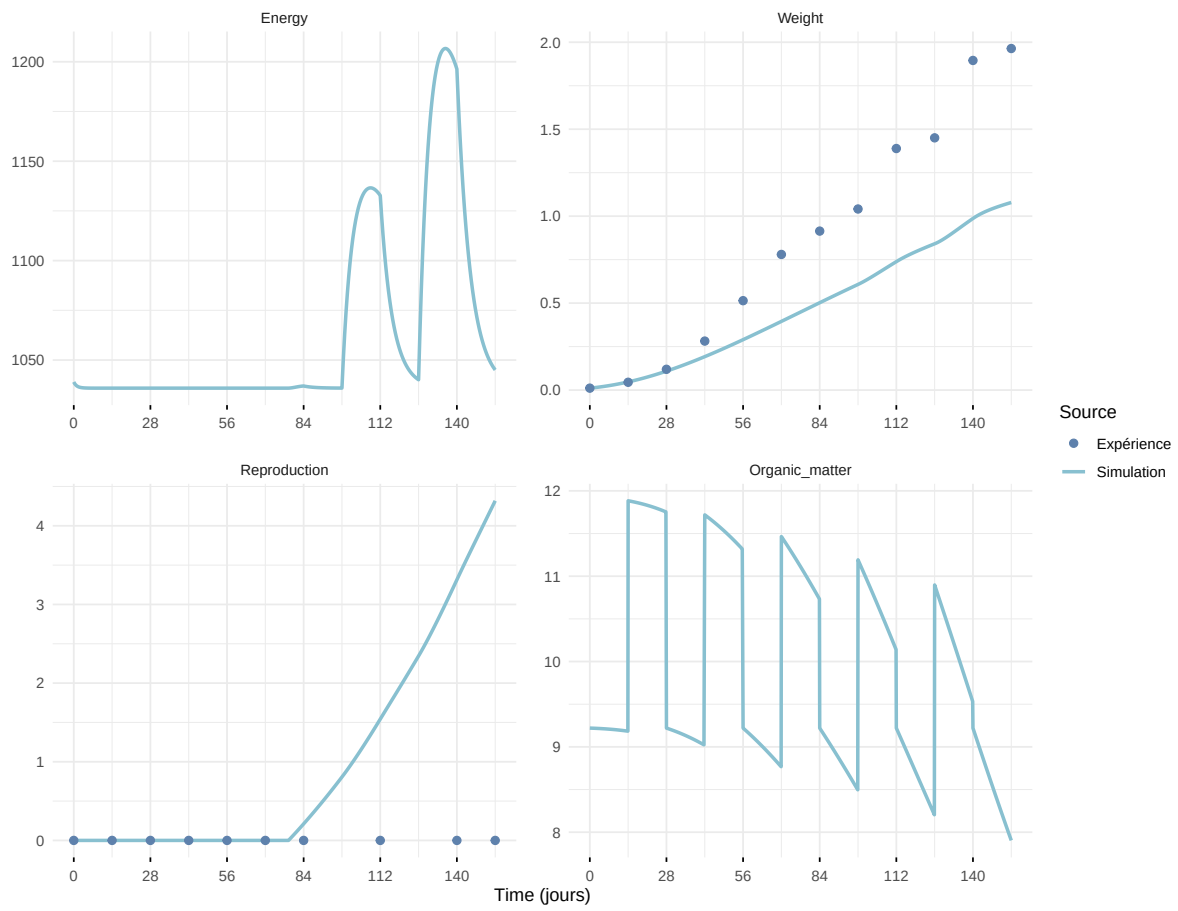
Expérience : Bart2019_XP5



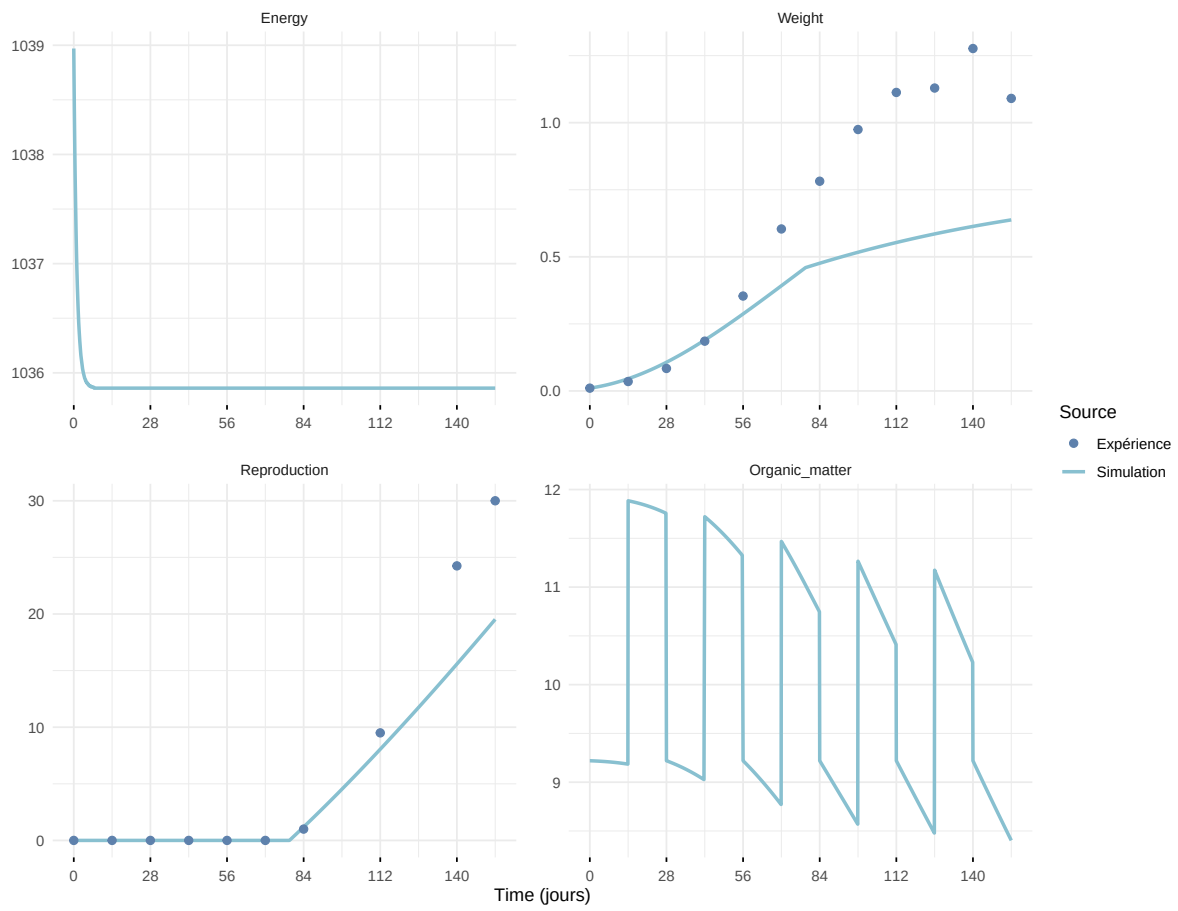
Expérience : Bart2020



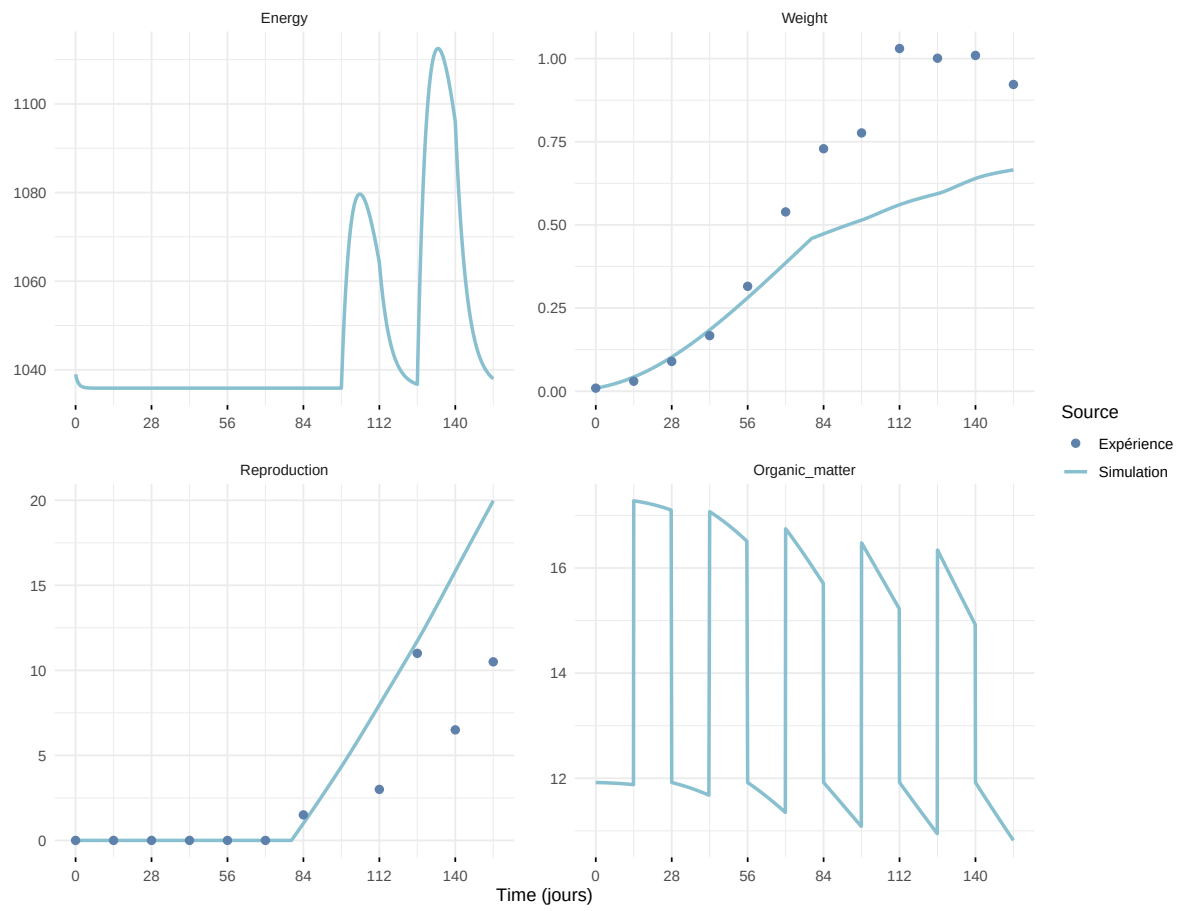
Expérience : Gollot2026_dens_D1



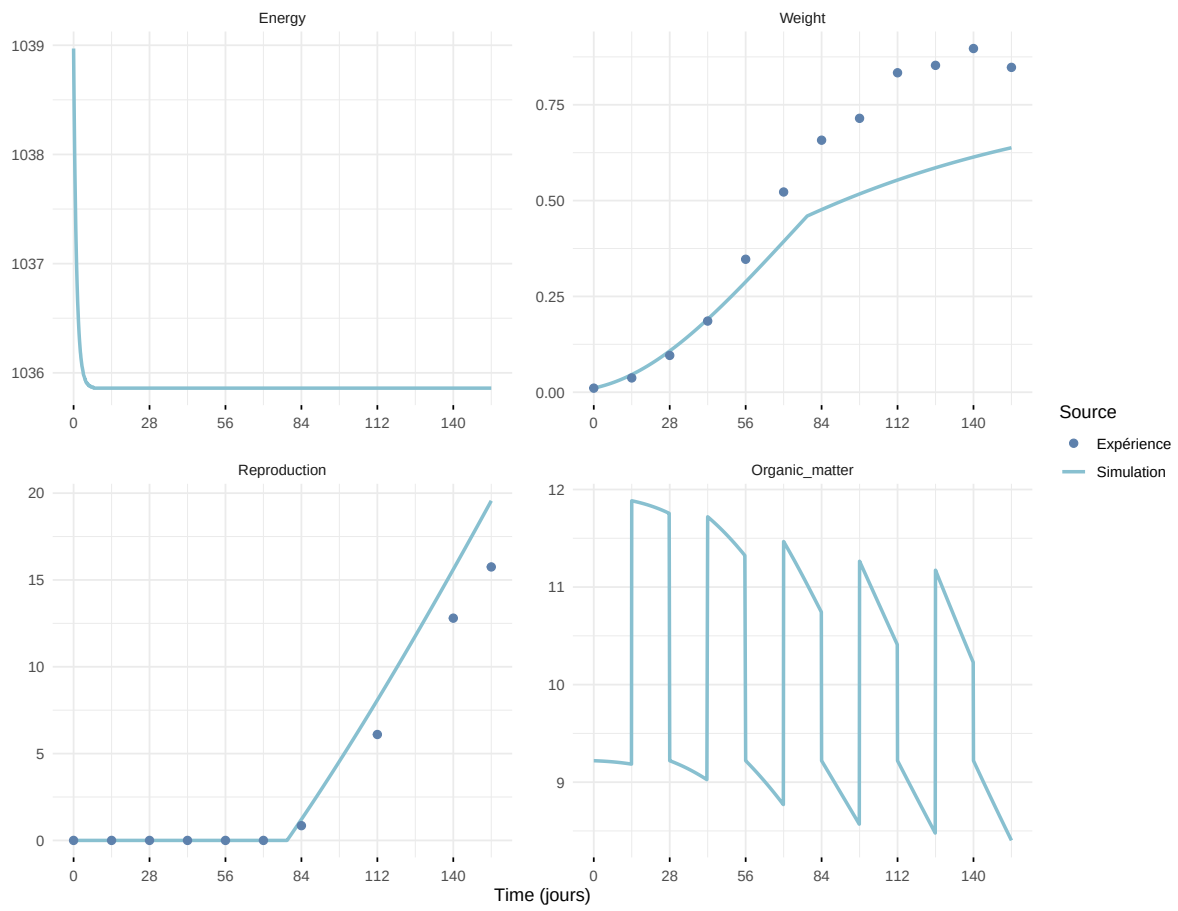
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

